

INTERNATIONAL SERIES IN PURE AND APPLIED MATHEMATICS



PRINCIPLES OF
MATHEMATICAL ANALYSIS

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CONTENTS

Preface	ix
Chapter 1 The Real and Complex Number Systems	1
Introduction	1
Ordered Sets	3
Fields	5
The Real Field	8
The Extended Real Number System	11
The Complex Field	12
Euclidean Spaces	16
Appendix	17
Exercises	21
Chapter 2 Basic Topology	24
Finite, Countable, and Uncountable Sets	24
Metric Spaces	30
Compact Sets	36
Perfect Sets	41

Connected Sets	42
Exercises	43
Chapter 3 Numerical Sequences and Series	47
Convergent Sequences	47
Subsequences	51
Cauchy Sequences	52
Upper and Lower Limits	55
Some Special Sequences	57
Series	58
Series of Nonnegative Terms	61
The Number e	63
The Root and Ratio Tests	65
Power Series	69
Summation by Parts	70
Absolute Convergence	71
Addition and Multiplication of Series	72
Rearrangements	75
Exercises	78
Chapter 4 Continuity	83
Limits of Functions	83
Continuous Functions	85
Continuity and Compactness	89
Continuity and Connectedness	93
Discontinuities	94
Monotonic Functions	95
Infinite Limits and Limits at Infinity	97
Exercises	98
Chapter 5 Differentiation	103
The Derivative of a Real Function	103
Mean Value Theorems	107
The Continuity of Derivatives	108
L'Hospital's Rule	109
Derivatives of Higher Order	110
Taylor's Theorem	110
Differentiation of Vector-valued Functions	111
Exercises	114

Chapter 6	The Riemann-Stieltjes Integral	120
	Definition and Existence of the Integral	120
	Properties of the Integral	128
	Integration and Differentiation	133
	Integration of Vector-valued Functions	135
	Rectifiable Curves	136
	Exercises	138
Chapter 7	Sequences and Series of Functions.	143
	Discussion of Main Problem	143
	Uniform Convergence	147
	Uniform Convergence and Continuity	149
	Uniform Convergence and Integration	151
	Uniform Convergence and Differentiation	152
	Equicontinuous Families of Functions	154
	The Stone-Weierstrass Theorem	159
	Exercises	165
Chapter 8	Some Special Functions	172
	Power Series	172
	The Exponential and Logarithmic Functions	178
	The Trigonometric Functions	182
	The Algebraic Completeness of the Complex Field	184
	Fourier Series	185
	The Gamma Function	192
	Exercises	196
Chapter 9	Functions of Several Variables	204
	Linear Transformations	204
	Differentiation	211
	The Contraction Principle	220
	The Inverse Function Theorem	221
	The Implicit Function Theorem	223
	The Rank Theorem	228
	Determinants	231
	Derivatives of Higher Order	235
	Differentiation of Integrals	236
	Exercises	239
Chapter 10	Integration of Differential Forms	245
	Integration	245

Primitive Mappings	248
Partitions of Unity	251
Change of Variables	252
Differential Forms	253
Simplexes and Chains	266
Stokes' Theorem	273
Closed Forms and Exact Forms	275
Vector Analysis	280
Exercises	288
Chapter 11 The Lebesgue Theory	300
Set Functions	300
Construction of the Lebesgue Measure	302
Measure Spaces	310
Measurable Functions	310
Simple Functions	313
Integration	314
Comparison with the Riemann Integral	322
Integration of Complex Functions	325
Functions of Class $\mathcal{L}^2$	325
Exercises	332
Bibliography	335
List of Special Symbols	337
Index	339

PREFACE

This book is intended to serve as a text for the course in analysis that is usually taken by advanced undergraduates or by first-year students who study mathematics.

The present edition covers essentially the same topics as the second one, with some additions, a few minor omissions, and considerable rearrangement. I hope that these changes will make the material more accessible and more attractive to the students who take such a course.

Experience has convinced me that it is pedagogically unsound (though logically correct) to start off with the construction of the real numbers from the rational ones. At the beginning, most students simply fail to appreciate the need for doing this. Accordingly, the real number system is introduced as an ordered field with the least-upper-bound property, and a few interesting applications of this property are quickly made. However, Dedekind's construction is not omitted. It is now in an Appendix to Chapter 1, where it may be studied and enjoyed whenever the time seems ripe.

The material on functions of several variables is almost completely rewritten, with many details filled in, and with more examples and more motivation. The proof of the inverse function theorem—the key item in Chapter 9—is

simplified by means of the fixed point theorem about contraction mappings. Differential forms are discussed in much greater detail. Several applications of Stokes' theorem are included.

As regards other changes, the chapter on the Riemann-Stieltjes integral has been trimmed a bit, a short do-it-yourself section on the gamma function has been added to Chapter 8, and there is a large number of new exercises, most of them with fairly detailed hints.

I have also included several references to articles appearing in the *American Mathematical Monthly* and in *Mathematics Magazine*, in the hope that students will develop the habit of looking into the journal literature. Most of these references were kindly supplied by R. B. Burckel.

Over the years, many people, students as well as teachers, have sent me corrections, criticisms, and other comments concerning the previous editions of this book. I have appreciated these, and I take this opportunity to express my sincere thanks to all who have written me.

WALTER RUDIN

THE REAL AND COMPLEX NUMBER SYSTEMS

INTRODUCTION

A satisfactory discussion of the main concepts of analysis (such as convergence, continuity, differentiation, and integration) must be based on an accurately defined number concept. We shall not, however, enter into any discussion of the axioms that govern the arithmetic of the integers, but assume familiarity with the rational numbers (i.e., the numbers of the form m/n , where m and n are integers and $n \neq 0$).

The rational number system is inadequate for many purposes, both as a field and as an ordered set. (These terms will be defined in Secs. 1.6 and 1.12.) For instance, there is no rational p such that $p^2 = 2$. (We shall prove this presently.) This leads to the introduction of so-called “irrational numbers” which are often written as infinite decimal expansions and are considered to be “approximated” by the corresponding finite decimals. Thus the sequence

$$1, 1.4, 1.41, 1.414, 1.4142, \dots$$

“tends to $\sqrt{2}$.” But unless the irrational number $\sqrt{2}$ has been clearly defined, the question must arise: Just what is it that this sequence “tends to”?

This sort of question can be answered as soon as the so-called “real number system” is constructed.

1.1 Example We now show that the equation

$$(1) \quad p^2 = 2$$

is not satisfied by any rational p . If there were such a p , we could write $p = m/n$ where m and n are integers that are not both even. Let us assume this is done. Then (1) implies

$$(2) \quad m^2 = 2n^2,$$

This shows that m^2 is even. Hence m is even (if m were odd, m^2 would be odd), and so m^2 is divisible by 4. It follows that the right side of (2) is divisible by 4, so that n^2 is even, which implies that n is even.

The assumption that (1) holds thus leads to the conclusion that both m and n are even, contrary to our choice of m and n . Hence (1) is impossible for rational p .

We now examine this situation a little more closely. Let A be the set of all positive rationals p such that $p^2 < 2$ and let B consist of all positive rationals p such that $p^2 > 2$. We shall show that A contains no largest number and B contains no smallest.

More explicitly, for every p in A we can find a rational q in A such that $p < q$, and for every p in B we can find a rational q in B such that $q < p$.

To do this, we associate with each rational $p > 0$ the number

$$(3) \quad q = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2}.$$

Then

$$(4) \quad q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2}.$$

If p is in A then $p^2 - 2 < 0$, (3) shows that $q > p$, and (4) shows that $q^2 < 2$. Thus q is in A .

If p is in B then $p^2 - 2 > 0$, (3) shows that $0 < q < p$, and (4) shows that $q^2 > 2$. Thus q is in B .

1.2 Remark The purpose of the above discussion has been to show that the rational number system has certain gaps, in spite of the fact that between any two rationals there is another: If $r < s$ then $r < (r + s)/2 < s$. The real number system fills these gaps. This is the principal reason for the fundamental role which it plays in analysis.

In order to elucidate its structure, as well as that of the complex numbers, we start with a brief discussion of the general concepts of *ordered set* and *field*.

Here is some of the standard set-theoretic terminology that will be used throughout this book.

1.3 Definitions If A is any set (whose elements may be numbers or any other objects), we write $x \in A$ to indicate that x is a member (or an element) of A .

If x is not a member of A , we write: $x \notin A$.

The set which contains no element will be called the *empty set*. If a set has at least one element, it is called *nonempty*.

If A and B are sets, and if every element of A is an element of B , we say that A is a subset of B , and write $A \subset B$, or $B \supset A$. If, in addition, there is an element of B which is not in A , then A is said to be a *proper* subset of B . Note that $A \subset A$ for every set A .

If $A \subset B$ and $B \subset A$, we write $A = B$. Otherwise $A \neq B$.

1.4 Definition Throughout Chap. 1, the set of all rational numbers will be denoted by Q .

ORDERED SETS

1.5 Definition Let S be a set. An *order* on S is a relation, denoted by $<$, with the following two properties:

(i) If $x \in S$ and $y \in S$ then one and only one of the statements

$$x < y, \quad x = y, \quad y < x$$

is true.

(ii) If $x, y, z \in S$, if $x < y$ and $y < z$, then $x < z$.

The statement " $x < y$ " may be read as " x is less than y " or " x is smaller than y " or " x precedes y ".

It is often convenient to write $y > x$ in place of $x < y$.

The notation $x \leq y$ indicates that $x < y$ or $x = y$, without specifying which of these two is to hold. In other words, $x \leq y$ is the negation of $x > y$.

1.6 Definition An *ordered set* is a set S in which an order is defined.

For example, Q is an ordered set if $r < s$ is defined to mean that $s - r$ is a positive rational number.

1.7 Definition Suppose S is an ordered set, and $E \subset S$. If there exists a $\beta \in S$ such that $x \leq \beta$ for every $x \in E$, we say that E is *bounded above*, and call β an *upper bound* of E .

Lower bounds are defined in the same way (with $\geq$ in place of $\leq$).

1.8 Definition Suppose S is an ordered set, $E \subset S$, and E is bounded above. Suppose there exists an $\alpha \in S$ with the following properties:

- (i) α is an upper bound of E .
- (ii) If $\gamma < \alpha$ then γ is not an upper bound of E .

Then α is called the *least upper bound* of E [that there is at most one such α is clear from (ii)] or the *supremum* of E , and we write

$$\alpha = \sup E.$$

The *greatest lower bound*, or *infimum*, of a set E which is bounded below is defined in the same manner: The statement

$$\alpha = \inf E$$

means that α is a lower bound of E and that no β with $\beta > \alpha$ is a lower bound of E .

1.9 Examples

(a) Consider the sets A and B of Example 1.1 as subsets of the ordered set Q . The set A is bounded above. In fact, the upper bounds of A are exactly the members of B . Since B contains no smallest member, A has no least upper bound in Q .

Similarly, B is bounded below: The set of all lower bounds of B consists of A and of all $r \in Q$ with $r \leq 0$. Since A has no largest member, B has no greatest lower bound in Q .

(b) If $\alpha = \sup E$ exists, then α may or may not be a member of E . For instance, let E_1 be the set of all $r \in Q$ with $r < 0$. Let E_2 be the set of all $r \in Q$ with $r \leq 0$. Then

$$\sup E_1 = \sup E_2 = 0,$$

and $0 \notin E_1$, $0 \in E_2$.

(c) Let E consist of all numbers $1/n$, where $n = 1, 2, 3, \dots$. Then $\sup E = 1$, which is in E , and $\inf E = 0$, which is not in E .

1.10 Definition An ordered set S is said to have the *least-upper-bound property* if the following is true:

If $E \subset S$, E is not empty, and E is bounded above, then $\sup E$ exists in S .

Example 1.9(a) shows that Q does not have the least-upper-bound property.

We shall now show that there is a close relation between greatest lower bounds and least upper bounds, and that every ordered set with the least-upper-bound property also has the greatest-lower-bound property.

1.11 Theorem Suppose S is an ordered set with the least-upper-bound property, $B \subset S$, B is not empty, and B is bounded below. Let L be the set of all lower bounds of B . Then

$$\alpha = \sup L$$

exists in S , and $\alpha = \inf B$.

In particular, $\inf B$ exists in S .

Proof Since B is bounded below, L is not empty. Since L consists of exactly those $y \in S$ which satisfy the inequality $y \leq x$ for every $x \in B$, we see that every $x \in B$ is an upper bound of L . Thus L is bounded above. Our hypothesis about S implies therefore that L has a supremum in S ; call it α .

If $\gamma < \alpha$ then (see Definition 1.8) γ is not an upper bound of L , hence $\gamma \notin B$. It follows that $\alpha \leq x$ for every $x \in B$. Thus $\alpha \in L$.

If $\alpha < \beta$ then $\beta \notin L$, since α is an upper bound of L .

We have shown that $\alpha \in L$ but $\beta \notin L$ if $\beta > \alpha$. In other words, α is a lower bound of B , but β is not if $\beta > \alpha$. This means that $\alpha = \inf B$.

FIELDS

1.12 Definition A *field* is a set F with two operations, called *addition* and *multiplication*, which satisfy the following so-called “field axioms” (A), (M), and (D):

(A) Axioms for addition

- (A1) If $x \in F$ and $y \in F$, then their sum $x + y$ is in F .
- (A2) Addition is commutative: $x + y = y + x$ for all $x, y \in F$.
- (A3) Addition is associative: $(x + y) + z = x + (y + z)$ for all $x, y, z \in F$.
- (A4) F contains an element 0 such that $0 + x = x$ for every $x \in F$.
- (A5) To every $x \in F$ corresponds an element $-x \in F$ such that

$$x + (-x) = 0.$$

(M) Axioms for multiplication

- (M1) If $x \in F$ and $y \in F$, then their product xy is in F .
- (M2) Multiplication is commutative: $xy = yx$ for all $x, y \in F$.
- (M3) Multiplication is associative: $(xy)z = x(yz)$ for all $x, y, z \in F$.
- (M4) F contains an element $1 \neq 0$ such that $1x = x$ for every $x \in F$.
- (M5) If $x \in F$ and $x \neq 0$ then there exists an element $1/x \in F$ such that

$$x \cdot (1/x) = 1.$$